

Beginner

1. Create a set with numbers from 1 to 10 and print it.
 2. Take 5 numbers as input from the user and store them in a set.
 3. Find the length of a set without using len().
 4. Check if an element exists in a set.
 5. Add an element to a set.
 6. Remove an element from a set.
 7. Clear all elements from a set.
 8. Iterate through a set and print each element.
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Easy

9. Find the union of two sets.

A = {1,2,3,4}

B = {3,4,5,6}

10. Find the intersection of two sets.
11. Find the difference A - B.
12. Find the symmetric difference of two sets.
13. Check if two sets are equal.
14. Check whether one set is a subset of another.
15. Check whether one set is a superset of another.
16. Count the number of unique elements in a list using a set.

l = [1,2,2,3,4,4,5,5]

Medium

17. Remove duplicates from a list using a set.
18. Find common elements between two lists using sets.

l1 = [1,2,3,4,5]

l2 = [4,5,6,7,8]

19. Find all unique vowels present in a string.

"programming"

20. Check whether two strings are anagrams using sets and other logic.

listen

silent

21. Find the missing numbers between two sets.

A = {1,2,3,4,5,6,7,8,9}

B = {2,4,6,8}

22. Convert a string into a set of unique characters.

23. Find all characters that appear in both strings.

s1 = "python"

s2 = "typhoon"

24. Find elements that appear in exactly one of two sets.

Hard

25. Given a list of names, find all unique first letters.

["Amit", "Ankit", "Rahul", "Rohan", "Neha"]

26. Find the first repeated element in a list using a set.

[5,3,1,4,3,2,1]

27. Check whether a list contains duplicates using a set.

28. Find the longest word in a sentence that has all unique characters.

29. Given two sentences, find all common words.

30. Find all unique words in a paragraph.

31. Find pairs of numbers whose sum is equal to a target value using a set.

nums = [2,7,11,15,3,6]

target = 9

32. Given a list of integers, find whether any two numbers differ by exactly k.

nums = [1,5,3,4,2]

k = 2

Interview Style

33. Find the second largest unique element in a list.

[5,2,8,8,1,9,9,4]

34. Find all duplicate elements in a list using sets.

35. Check if two lists contain exactly the same elements (order doesn't matter).

36. Find the intersection of three sets.

37. Find the number of distinct words in a sentence.

38. Determine whether every character in a string is unique.

39. Find the largest consecutive sequence length.

[100,4,200,1,3,2]

40. Group words that contain the same set of characters.

["bat", "tab", "cat", "act", "tac"]